

Abhishek Bharadwaj

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Career Objective

Electronics and Instrumentation Engineering student with hands-on experience in STM32 firmware development, Embedded C, and sensor interfacing, seeking an Embedded Systems or Firmware Engineering role..

Education

Sri Jayachamarajendra College of Engineering (JSS-STU), Mysuru

Bachelor of Engineering in Electronics and Instrumentation Engineering

Nov 2022 – Present

Technical Skills

- **Programming Languages:** C, C++, Python
- **Microcontrollers & Platforms:** STM32F411CE (ARM Cortex-M4), Raspberry Pi Zero 2 W
- **Communication Protocols:** SPI, I2C, UART/USART, 1-Wire
- **Embedded Systems:** Register-level programming, GPIO debugging, sensor interfacing, circuit prototyping, soldering
- **Testing & Debugging Tools:** Oscilloscope, Digital Multimeter, ST-LINK SWD Debugging
- **Software & Tools:** STM32CubeIDE, VS Code, Git, GitHub, Linux (Basics)

Projects

Bare-Metal Driver Development for STM32F411CE (Black Pill)

Dec 2025

GitHub: STM32F411xx_Driver

- Developed register-level GPIO drivers in Embedded C for STM32F411CE microcontroller.
- Implemented GPIO configuration, pull-up/pull-down settings, alternate function mapping, and atomic GPIO operations using BSRR registers.
- Designed EXTI interrupt handling with NVIC priority configuration and interrupt management.
- Strengthened understanding of ARM Cortex-M4 architecture and memory-mapped peripheral programming.
- The project is still being updated with other peripheral drivers such as SPI, UART/USART, I2C.

FreeRTOS UART Command Console on STM32F411

Oct 2025 – Nov 2025

- Developed a command-line interface on STM32F411 using FreeRTOS and UART communication.
- Implemented multitasking architecture with dedicated UART reception, command processing, and system monitoring tasks.
- Supported runtime commands for LED control, system status monitoring, and diagnostic reporting.

Neonatal Monitoring System

Jun 2025 – Jul 2025

- Developed a real-time neonatal monitoring system using Raspberry Pi Zero 2 W.
- Interfaced MAX30102, MPU6050, and DS18B20 sensors using I2C and 1-Wire protocols.
- Processed heart rate, SpO2, temperature, and motion data for continuous monitoring applications.

Training & Certifications

- Embedded Systems Training – Internshala (Sep 2025 – Nov 2025)
- Mastering Microcontroller and Embedded Driver Development – FastBit Academy (Udemy, Ongoing)

Extra-Curricular Activities

Student Member, IEEE Student Branch, SJCE

2022 – 2024